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CYBERSECURITY INCIDENT SIMULATION AND ANALYSIS REPORT

Simulated cybersecurity incident analysis performed using Windows logs, PowerShell activity, process inspection, and network forensics to identify suspicious behavior, indicators of compromise (IOCs), and recommended containment actions.

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Introduction:

Goal:

Simulate a small cybersecurity incident and perform basic forensic analysis using Windows logs, PowerShell activity, running processes, active network connections, and authentication events to identify suspicious behavior and potential Indicators of Compromise (IOCs).

Requirements:

- Analyze suspicious activity from Windows Event Logs and PowerShell history
- Identify possible Indicators of Compromise (IOCs)
- Investigate failed authentication attempts and reconnaissance activity
- Examine running processes, active services, and network connections
- Document forensic findings and containment actions
- Provide security recommendations based on observed activities

Deliverables:

- Detailed Cybersecurity Incident Analysis Report (PDF/DOC)
- List of detected anomalies and Indicators of Compromise (IOCs)
- Screenshots or outputs of PowerShell commands, Event Viewer logs, and network analysis
- Documentation of response steps and containment measures
- Final conclusions and security recommendations

Executive Summary:

A simulated cybersecurity incident investigation was conducted on a Windows-based workstation to analyze suspicious activities and identify potential Indicators of Compromise (IOCs). The investigation focused on forensic evidence collected from:

- PowerShell command history
- Authentication logs
- Running processes
- Active network connections
- Windows Event Viewer logs
- System services

The analysis identified:

- Encoded PowerShell execution
- Failed login attempts
- Reconnaissance commands
- Suspicious process activity
- Unusual network connection states

The simulation successfully demonstrated common attacker techniques including reconnaissance activity, encoded PowerShell execution, authentication failures, and suspicious network behavior. Evidence gathered during the investigation resembled tactics commonly associated with early-stage cyber intrusion attempts.

Although no confirmed malware compromise was identified, the presence of suspicious encoded PowerShell activity and an unusual process name highlighted the importance of continuous monitoring and endpoint investigation.

System Information:

Parameter	Value
Hostname	BT0959
Username	KIIT
Operating System	Windows
IPv4 Address	192.168.31.49
Default Gateway	192.168.31.1

Incident Scenario:

The cybersecurity simulation was designed to imitate common attacker behavior on a Windows endpoint. Activities performed during the simulation included:

- Executing reconnaissance commands
- Running encoded PowerShell commands
- Reviewing active processes
- Monitoring network connections
- Generating failed login attempts
- Investigating Windows event logs
- Inspecting running security services

The objective was to understand how suspicious activity appears within system logs and forensic artifacts and to practice identifying possible indicators of compromise.

Investigation Methodology:

The forensic investigation followed a structured incident response methodology consisting of the following phases:

Phase 1 — Evidence Collection

System artifacts and logs were collected using built-in Windows administrative tools and PowerShell commands.

Phase 2 — Log Analysis

Authentication logs, PowerShell history, and system events were reviewed to identify suspicious behavior.

Phase 3 — Process and Service Inspection

Running processes and services were analyzed to identify abnormal or potentially malicious activity.

Phase 4 — Network Analysis

Active network sessions and listening ports were reviewed to identify external communications and suspicious connection states.

Phase 5 — IOC Identification

Potential indicators of compromise were extracted and documented.

Tools and Utilities Used:

Tool / Command	Purpose
PowerShell	Command execution and forensic analysis
eventvwr.msc	Windows Event Viewer analysis
services.msc	Service inspection
tasklist	Running process enumeration
netstat -ano	Active network connection analysis
whoami	User identification
ipconfig	Network configuration analysis
net user	Local user enumeration
Get-History	PowerShell history review

Suspicious Log Activity Analysis:

1(a).PowerShell History Analysis:

Command Used: “Get-History”

Collected PowerShell History

ID	Command
1	whoami
2	ipconfig
3	tasklist
4	net user
5	powershell -EncodedCommand ZQBjAGgAbwAgACIASABhAGMAawBLAGQAAlgA=

Analysis

The executed commands demonstrate typical reconnaissance activity used by attackers after gaining access to a system.

1(b).Encoded PowerShell Execution:

Observed Command:

“powershell -EncodedCommand
ZQBjAGgAbwAgACIASABhAGMAawBLAGQAAlgA=”

Decoded Output:

“echo "Hacked"”

Analysis:

Encoded PowerShell execution is frequently associated with malicious activity because attackers often use Base64-encoded commands to:

- Obfuscate malicious payloads
- Evade security monitoring
- Bypass antivirus detections
- Hide attacker intentions from logs

Although the decoded payload in this simulation was harmless, the execution technique closely resembles real-world attacker behavior.

1(c).Reconnaissance Activity Analysis:

The following reconnaissance commands were executed during the simulation:

Command	Purpose
whoami	Identify current logged-in user
ipconfig	Retrieve network configuration
tasklist	Enumerate running processes
net user	Enumerate local user accounts

Command Output Analysis

whoami: “bt0959\kiit”

This confirms the active user context under which commands were executed.

ipconfig Findings

The system contained:

- Active Wi-Fi connection
- IPv4 address: 192.168.31.49
- Multiple OpenVPN virtual adapters
- Cloudflare WARP-related network activity
- IPv6 connectivity enabled

Security Observation

The presence of multiple VPN adapters indicates active tunneling software usage, which can complicate forensic attribution and traffic inspection.

net user Findings

Local accounts identified:

- Administrator
- DefaultAccount
- Guest
- KIIT
- WDAGUtilityAccount

Analysis

Enumeration of user accounts is commonly performed during post-compromise reconnaissance to identify privileged accounts and possible lateral movement targets.

2.Windows Security logs analysis:

(a).Failed Login Attempt Analysis:

Event Source: Windows Security Logs
Access Method: “eventvwr.msc”
Event Filter Used: “Event ID 4625”

Extracted Event Details:

Field	Value
Event ID	4625
Target Username	KIIT
Status Code	0xc000006d
SubStatus Code	0xc000006a
Logon Type	2
Source IP Address	127.0.0.1
Process Name	C:\Windows\System32\svchost.exe
Authentication Package	Negotiate
Workstation Name	BT0959

Event Interpretation:

Status Code – 0xc000006d
Meaning:
Authentication failure.
SubStatus Code – 0xc000006a
Meaning:
Incorrect password provided.
Logon Type – 2
Meaning:
Interactive local logon attempt.

Analysis:

Status Code – 0xc000006d
Meaning:
Authentication failure.
SubStatus Code – 0xc000006a
Meaning:
Incorrect password provided.
Logon Type – 2The failed authentication attempt originated locally from the system itself using the loopback address 127.0.0.1.
Possible explanations include:

- Incorrect password entry
- Credential guessing attempt
- Local brute-force simulation
- Unauthorized login attempt

Repeated occurrences of Event ID 4625 may indicate credential abuse attempts and should be monitored closely.
Meaning:
Interactive local logon attempt.

3(a).Running Process Analysis:

Command Used: “tasklist”

Notable Processes Observed:

Process	Observation
powershell.exe	Multiple PowerShell sessions active
openvpnserver.exe	OpenVPN service active
warp-svc.exe	Cloudflare WARP service active
chrome.exe	Numerous browser processes
msedgewebview2.exe	Multiple embedded web components
oracle.exe	Database-related process
MsMpEng.exe	Microsoft Defender Antivirus active
tmlaridae.exe	Suspicious/unverified process

3(b).Suspicious Process Investigation:

Process Name: “tmlaridae.exe”

Analysis:

The process name appears unusual and does not resemble standard Windows system processes.

Characteristics that make the process suspicious include:

- Non-standard naming convention
- Unknown software association
- Lack of obvious vendor identification
- Executing under user session context

Recommended Verification Steps

- Verify executable path
- Check digital signatures
- Upload hash to malware scanning platforms
- Review parent-child process relationships
- Analyze persistence mechanisms

4.Network Connection Analysis:

Command Used: “netstat -ano”

Active Network Findings:

Listening Ports:

Port	Service
135	RPC Endpoint Mapper
445	SMB/File Sharing
5040	Windows Service
24563	Epic Games Launcher
49728	Oracle Database
59999	Application Service

External Connections Observed:

Connections were identified to infrastructure associated with:

- Google
- Microsoft
- Cloudflare
- Facebook
- Epic Games

Suspicious Connection States:

State	Observation:
CLOSE_WAIT	Improperly terminated sessions
FIN_WAIT_2	Delayed session closure

Analysis:

Persistent abnormal TCP states may indicate:

- Unstable applications
- Incomplete session termination
- Network misuse
- Potential command-and-control behavior

VPN and Tunneling Activity

VPN-Related Ports:

Port	Purpose
UDP 500	IPsec
UDP 4500	NAT Traversal/IPsec

VPN Services Identified:

- OpenVPN
- Cloudflare WARP

Analysis:

The system was actively using VPN and tunneling applications, which encrypted outbound communications and introduced additional network interfaces into the environment.

5. Network Connection Analysis:

Tool Use: “services.msc”

Security Services Observed:

Service	Status
Windows Defender	Running
Microsoft Defender Antivirus	Running
OpenVPN Services	Running
Cloudflare WARP Services	Running
Trend Micro Components	Running

Analysis

Multiple endpoint protection and security services were operational during the investigation. The presence of active security tools reduced the likelihood of long-term malware persistence. However, attackers may still attempt to evade or disable these services during advanced attacks.

6. Event Viewer Investigation:

Tool Used: “eventvwr.msc”

Logs Investigated:

- Windows Security Logs
- Administrative Events
- PowerShell Logs
- Windows System Logs

Important Event IDs Reviewed:

Event ID	Description
4624	Successful login
4625	Failed login
4688	Process creation

Event Analysis:

Event ID 4625:

Used to identify failed authentication attempts.

Event ID 4688:

Useful for identifying suspicious process execution including PowerShell activity.

PowerShell Logs:

PowerShell operational logs help identify:

- Script execution
- Encoded command usage
- Administrative activity
- Suspicious automation behavior

7.Indicators of Compromise (IOCs):

IOC Type	Indicator
Encoded PowerShell	powershell -EncodedCommand
Failed Authentication	Event ID 4625
Incorrect Password	0xc000006a
Reconnaissance Commands	whoami, tasklist, net user
Suspicious Process	tmlaridae.exe
VPN Activity	OpenVPN and Cloudflare WARP
Suspicious TCP States	CLOSE_WAIT and FIN_WAIT_2
Multiple PowerShell Sessions	powershell.exe
Local Authentication Failure	127.0.0.1

8.Risk Assessment:

Threat Activity	Risk Level	Justification
Encoded PowerShell Execution	High	Common attacker obfuscation technique
Failed Login Attempts	Medium	Possible credential abuse activity
Reconnaissance Commands	Medium	Typical post-compromise enumeration
Suspicious Process	High	Unknown executable requires verification
Abnormal Network States	Medium	Possible unstable or malicious sessions
VPN Usage	Low	Legitimate but may obscure traffic

9.Containment and Response Actions:

Step 1 – Log Investigation:

Reviewed:

- PowerShell history
- Security logs
- Event Viewer logs
- Administrative events

Step 2 – Authentication Monitoring:

Investigated failed login attempts and reviewed authentication event details.

Step 3 – Process Verification:

Reviewed suspicious processes for possible malware indicators and unauthorized execution.

Step 4 – Network Inspection:

Analyzed active TCP and UDP sessions, listening ports, and external connections.

Step 5 – Security Validation:

Verified antivirus and endpoint protection services were operational and actively running.

10.List of Detected Anomalies:

Detected Anomaly	Description
Encoded PowerShell Execution	Suspicious encoded command execution
Failed Login Attempt	Incorrect password authentication failure
Reconnaissance Commands	Enumeration activity detected
Suspicious Process	Unverified process tmlaridae.exe
Unusual TCP States	CLOSE_WAIT and FIN_WAIT_2 observed
Multiple PowerShell Sessions	Several PowerShell instances running
VPN/Tunneling Activity	OpenVPN and WARP services active

11.Recommendations:

The following security improvements are recommended:

Authentication Security:

- Enable account lockout policies
- Monitor repeated Event ID 4625 activity
- Enforce strong password policies
- Implement multi-factor authentication where possible

PowerShell Security:

- Restrict encoded PowerShell execution
- Enable enhanced PowerShell logging
- Monitor suspicious script activity
- Enable Script Block Logging

Endpoint Monitoring:

- Investigate suspicious processes such as tmlaridae.exe
- Conduct regular antivirus scans
- Review startup entries and persistence mechanisms
- Monitor process creation events

Network Security:

- Regularly review open ports and listening services
- Monitor outbound encrypted traffic
- Audit VPN and tunneling software usage
- Investigate persistent abnormal TCP states

Logging and Detection:

- Centralize Windows logs using SIEM solutions
- Monitor Event IDs 4624, 4625, and 4688
- Enable advanced auditing policies
- Maintain regular forensic log reviews

12.Conclusion:

The simulated cybersecurity investigation successfully demonstrated several common attack techniques including:

- Failed authentication attempts
- PowerShell-based execution(Encoded PowerShell execution)
- Reconnaissance activity(System reconnaissance)
- Network enumeration
- Suspicious process activity

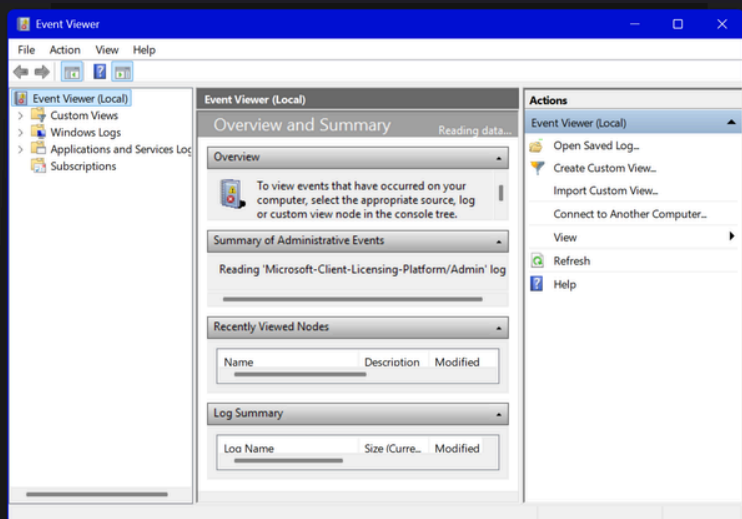
Although no confirmed malware infection was identified, the observed behaviors closely resemble techniques used during the early stages of real-world cyber intrusions.

The investigation highlights the importance of:

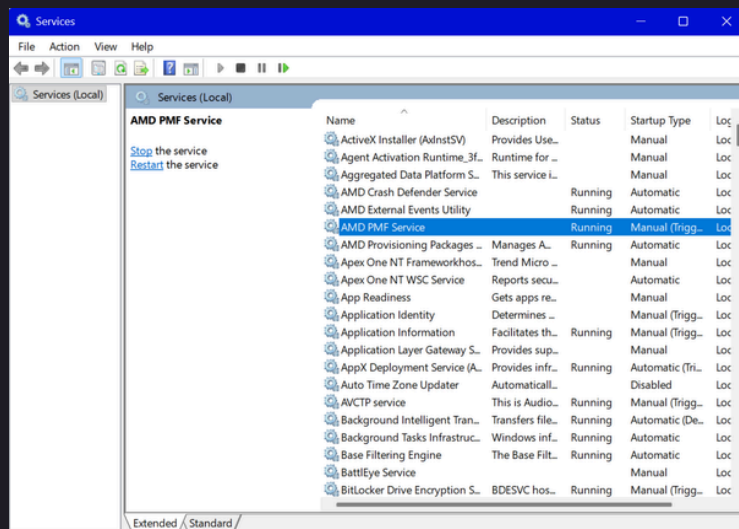
- Continuous monitoring
- Endpoint visibility
- PowerShell auditing
- Authentication analysis
- Network traffic inspection
- Security event correlation

Proper logging, endpoint protection, and regular forensic review remain essential components of an effective cybersecurity defense strategy.

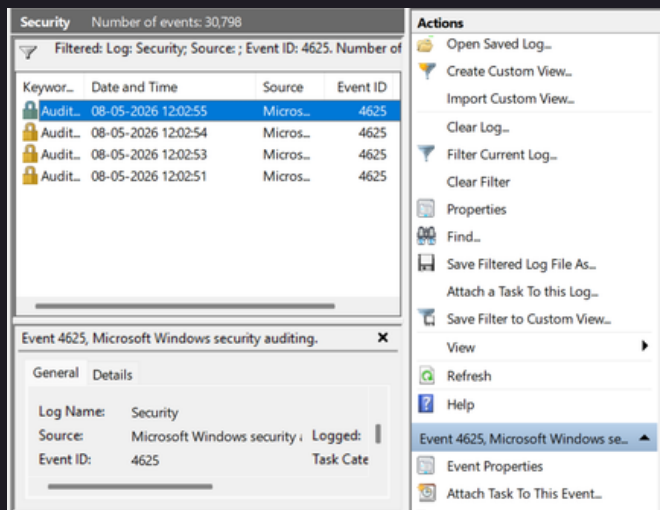
13.Screenshots:



Event Viewer screenshot



services.msc screenshot



Event log screenshot